

Hot Air Balloon: A Floating 3D Sky Scene

by

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Description

The project is an interactive 3D scene that will be developed in C++ and OpenGL, featuring a hot air balloon that launches from a countryside meadow and ascends into the sky. The scene includes a fully rigged balloon with a swaying basket and a flickering burner flame, a rotating windsock indicating wind direction, drifting clouds, a small flock of birds with flapping wings, and a day-to-night lighting cycle. An on-screen dashboard displays live altitude and wind-direction readings.

Objects in the Scene

- A hot air balloon will be built with an envelope (layered sphere/ellipsoid sections), a woven basket, connecting support ropes, and a burner unit with an animated flame.
- Two additional background balloons of varying sizes will drift at different heights to add depth and demonstrate independent transformation hierarchies.
- A launch platform will be constructed with a wooden base, a fence, and a rotating windsock mounted on a pole.
- A meadow/ground plane will be modeled with scattered low poly trees and a dirt path.
- Drifting clouds (flattened spheres) and a small flock of birds with hinged, flapping wings will populate the sky.
- A sun/moon disc will represent the primary light source, switching in a day/night cycle.
- A dashboard UI will display current altitude, wind speed/direction, and burner status.

Features

Transformations:

Translation (balloon ascent/drift, moving clouds/birds, shifting shadow); Rotation (swaying basket, flickering flame, spinning windsock, flapping wings); Scaling (differently sized balloons for depth, shrinking/growing shadow); Hierarchical (envelope → basket → burner → flame; bird body → wings).

Animation:

Continuous motion across all elements above: ascent, sway, flicker, drift, flap, spin, and shadow change.

Coordinate System and Projection:

Y-axis = altitude, X-axis = wind drift, Z-axis = depth; Perspective projection for the 3D scene, Orthographic projection for the dashboard HUD.

Camera:

Overview, Follow, Free-fly, and Basket POV modes; arrow keys move/orbit the free camera.

Lighting:

Ambient + Diffuse + Specular (base Phong model); Directional (sun/moon); Point (burner flame); Spotlight (launch-pad night light); two Area lights (sky fill + ground bounce).

Shadow:

A simple projected shadow of the balloon/basket on the ground, shrinking and shifting as it rises and drifts.

Keyboard Controls: 1–4 switch camera modes, Arrow keys move/orbit, Space pauses/resumes, L toggles day/night, F toggles the burner flame, R resets the scene, Esc exits.

